



魅力统计

Fascinating Statistics

第八章 因果推断

Chapter 8 Causal Inference

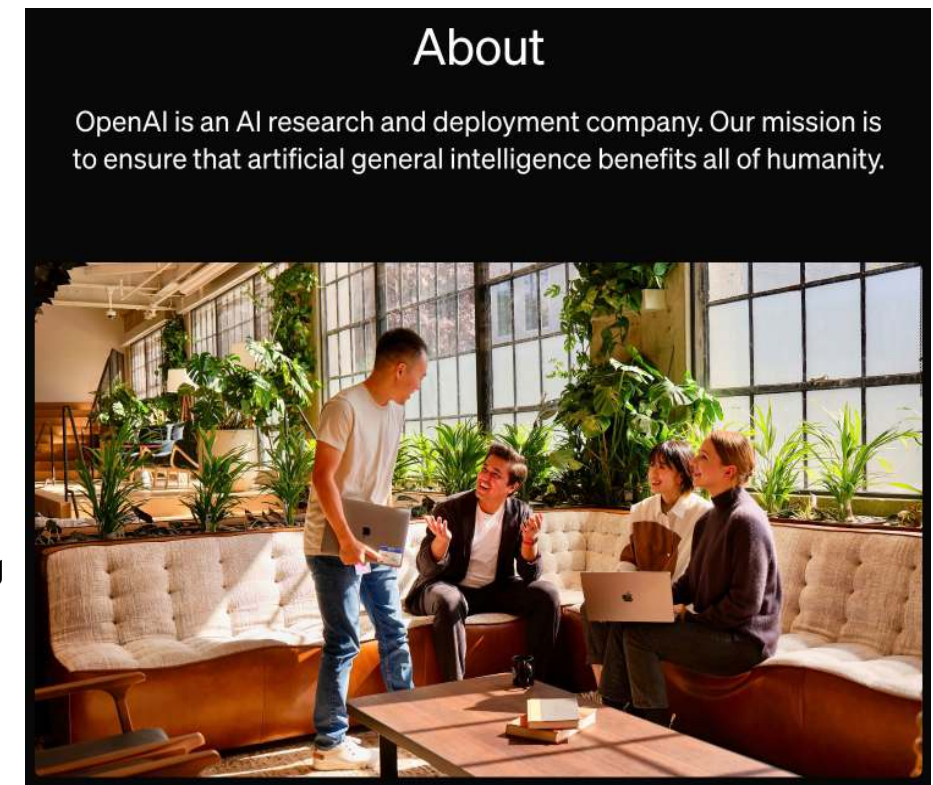
Chapter 8 Causal Inference

- 8.1 Introduction
- 8.2 Basic Concepts in Causal Inference



8.1 Introduction

- **AGI (artificial general intelligence, 通用人工智能)** is a research hotspot in the field of artificial intelligence, aiming to develop AI systems with autonomous self-control, rational self-understanding, and the ability to learn new skills.
- Unlike AI that focuses on solving specific tasks, AGI is endowed with broader **cognitive and reasoning abilities**, enabling it to learn, adapt, and perform tasks across multiple domains.
- The concept of AGI dates back to the mid-20th century, but it has gained renewed attention in recent years – largely because OpenAI explicitly includes it in its mission.
- In current AGI research, some scientist believe that building larger and more complex models could enhance AI's reasoning capabilities, suggesting that existing **LLMs (large language models)** might pave the way toward AGI.



来源: <https://openai.com/about/>



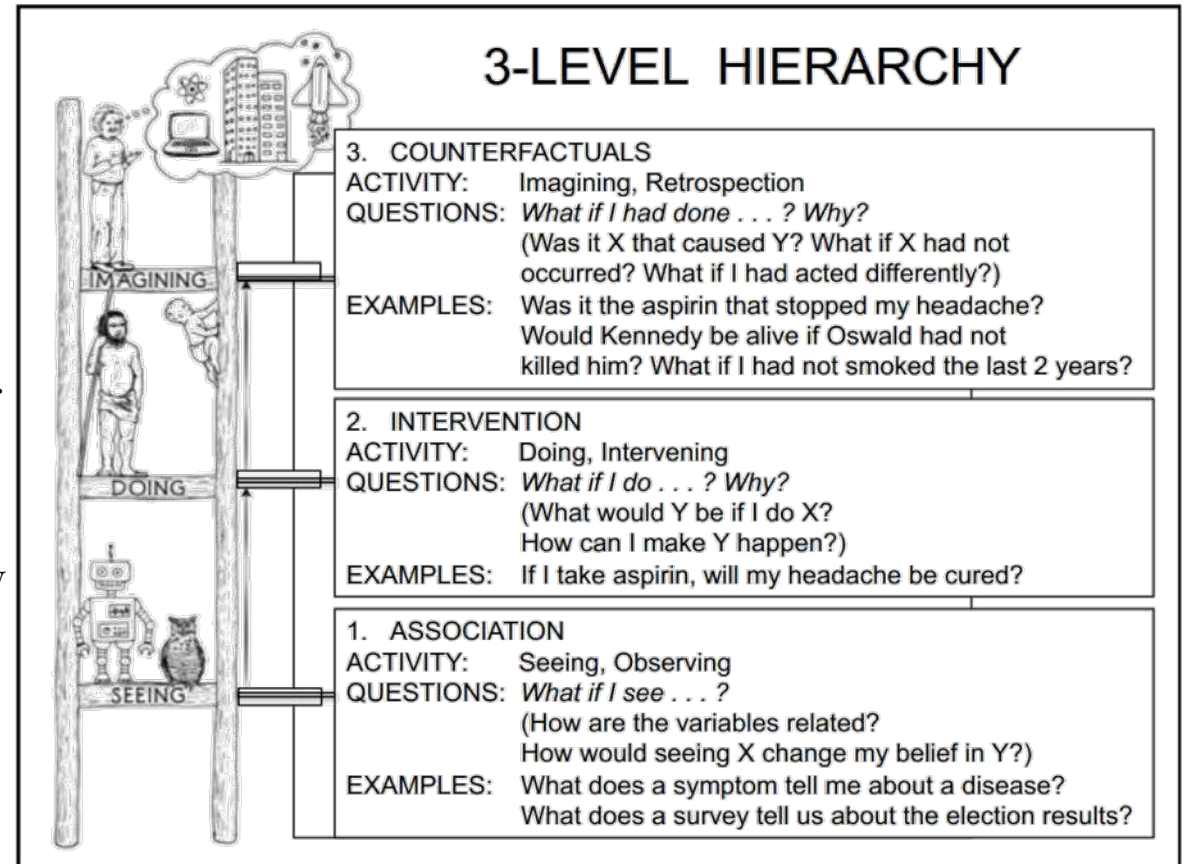
8.1 Introduction

- However, many researchers argue that while LLM excel at extracting information and patterns from data, they cannot generate novel causal insights.
- This explains why ChatGPT sometimes produces hallucinations or why Sora-generated videos may violate physical laws.
- They operate on a superficial approximation of reality, lacking genuine comprehension of how the world works.
- Although AI video generation models have seen substantial advancements in key areas, e.g., better **temporal consistency** and **adherence to basic physical laws**, they lack genuine understanding of physical laws.
- Causal inference plays a crucial role in AGI which allows AI to reason about the world in a way that humans do.



8.1 Introduction

- Turing award winner **Judea Pearl** outlines three ascending levels of causal reasoning in his book *The book of why*.
- The first level is **association** (关联), identifying patterns through passive observation.
 - Tools: Statistical/machine learning models.
 - Example: A supermarket sales manager may ask “what is the probability that customers who buy toothpaste will also buy dental floss?”
 - Limitations: Relies purely on detecting patterns in historical data without considering causality.

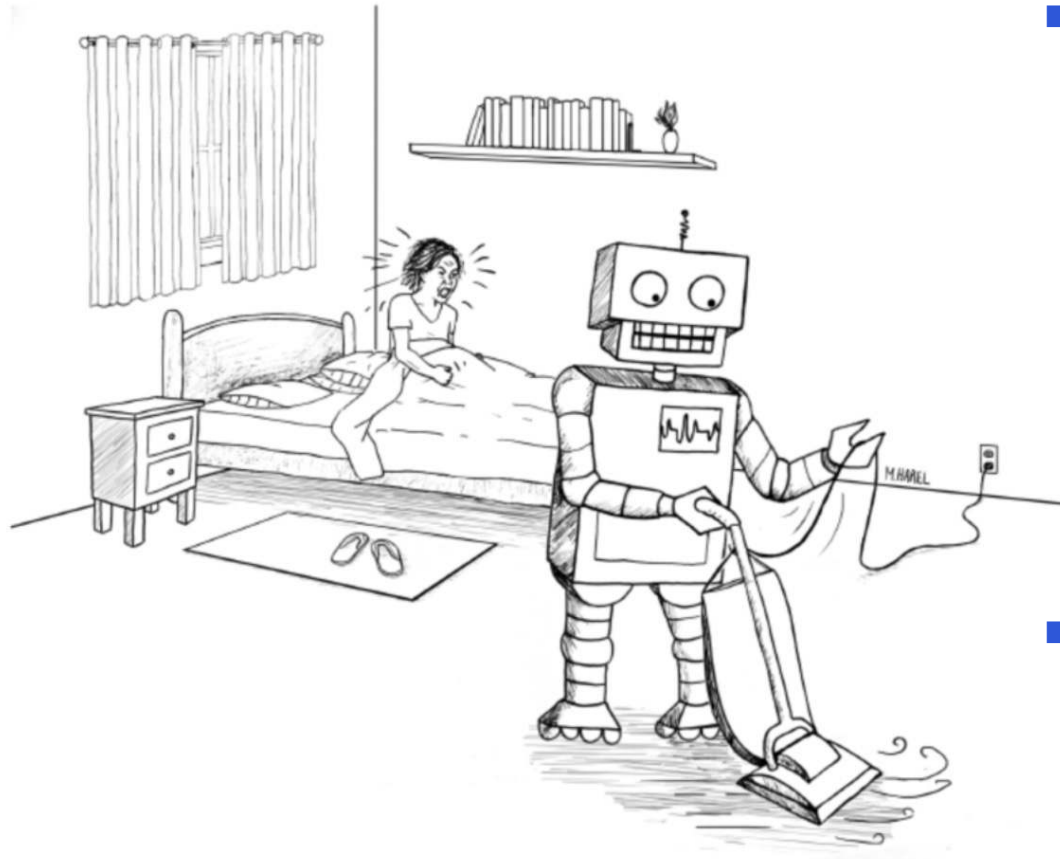


8.1 Introduction

- The second level is **intervention (干预)**, predicting outcomes of **deliberate actions**.
 - Tools: Randomized controlled trials, causal graph, do operator, etc.
 - Example: “If we double the price of toothpaste, how will dental floss sales be affected?”
 - Note: the difference between passive observation and active intervention.
- The third level is **counterfactuals (反事实)**, imagining and reflecting on what outcomes might occur in hypothetical scenarios if certain actions were taken.
 - Tools: Potential outcomes framework, structural causal models.
 - Example: “If I had doubled the price of toothpaste yesterday, what is the probability that customers who bought toothpaste would still have purchased it?”
 - Applications: Critical for accountability system (责任追究制度) in legal and medical scenarios.
- Pearl argues current AI excels at association but struggles with intervention and counterfactuals without explicit causal models.
- Each step along the ladder of causation enhances our ability to understand and shape the world, emphasizing the necessity of causal models for true intelligence, whether human or artificial.



8.1 Introduction



- Pearl said:
 - If, while I'm asleep, my home robot turns on the vacuum cleaner, I'll tell it: you shouldn't have woken me up.
 - What we expect of a truly intelligent robot is that it should understand what you and I take completely for granted that: the vacuum cleaner makes noise, that noise wakes people up, and that makes them unhappy.
- The example underscores Pearl's argument that AI must internalize causal mechanisms to act rationally and ethically.

8.1 Introduction

- In fact, relying solely on observational data without considering causal relationships can lead to misguided decisions.
- Simpson's paradox (辛普森悖论) serves as a classic example of this.
 - Suppose both Hospital A and Hospital B perform surgeries for a certain disease.
 - The following table records the recovery rates of patients after surgery at both hospitals, categorized by the patients' pre-operative conditions.
 - If you were a patient suffering from this disease, which hospital would you choose?

	Mild Case	Severe Case	Total
Hospital A	80% (720/900)	30% (30/100)	75% (750/1000)
Hospital B	90% (540/600)	40% (160/400)	70% (700/1000)



8.1 Introduction

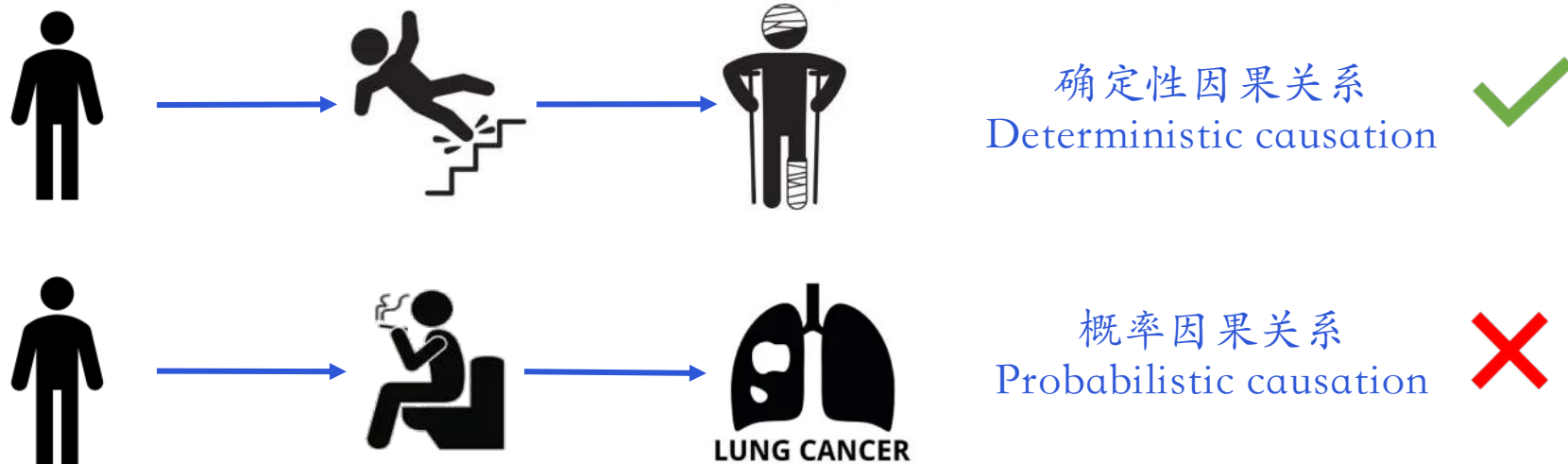
	Mild Case	Severe Case	Total
Hospital A	80% (720/900)	30% (30/100)	75% (750/1000)
Hospital B	90% (540/600)	40% (160/400)	70% (700/1000)

- We understand that the contradiction between overall and condition-specific recovery rates occurs because the proportions of mild and severe cases are different in the two hospitals.
- But does this suggest that we should choose Hospital B?
- The decision critically depends on the underlying causal structure of this problem.
 - If the patient's condition is the cause of his/her hospital choice (e.g., severe cases tend to choose Hospital B), then Hospital B would indeed be the better option.
 - Conversely, if the hospital choice is the cause of the patient's condition (e.g., Hospital B has limited resources leading to surgical delays that aggravate mild cases into severe ones), then Hospital A would actually be preferable.



8.1 Introduction

- Humans appear to be naturally skilled at causal reasoning – but not always.

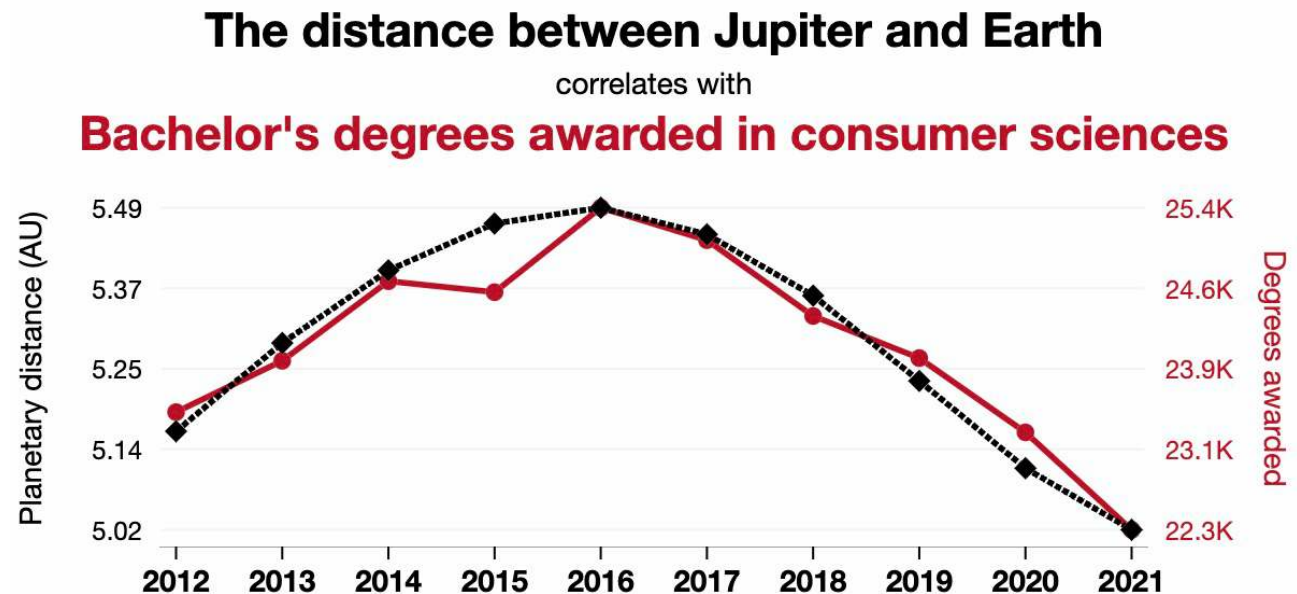
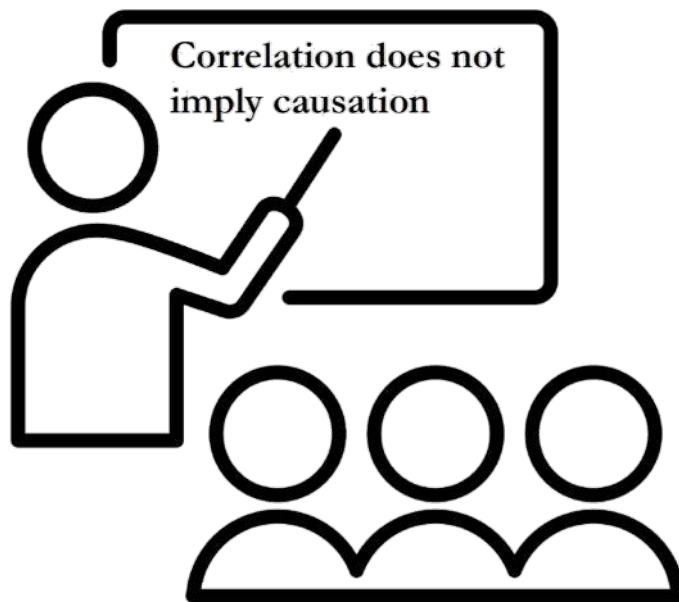


- While we can establish causation by **randomized controlled trial (RCT, 随机对照试验)**, many real-world scenarios preclude randomization.
- Much of epidemiology and social science research must inherently rely on **observational studies (观察性研究)**.
- How to extract causal knowledge from observational data is a key scientific problem.



8.1 Introduction

- When drawing conclusions through analysis based on observational studies, there is a popular statement
- People find many examples with high correlation but no casual relationship.



<https://www.tylervigen.com/spurious-correlations>



8.1 Introduction

- Therefore, many researchers try to avoid using causal language when stating their conclusions.

correlation
≠
causation

Use of Causal Language



Causal language (including use of terms such as effect and efficacy) should be used only for randomized clinical trials. For all other study designs (including meta-analyses of randomized clinical trials), methods and results should be described in terms of association or correlation and should avoid cause-and-effect wording. To read more about use of causal language, see the *AMA Manual of Style*.

- They talk around causality even when they think about it.
- This rhetorical ambiguity not only obscures meaning but permits contradictory practices, wherein causal disclaimers coexist with causal interpretations.
- Robust scientific methods for causal inference from observational studies is essential.

Association Link
Correlation Risk factor
Predictor

<https://jamanetwork.com/journals/jama/pages/instructions-for-authors>



8.1 Introduction

nature

Article | Published: 08 July 2020

Factors associated with COVID-19-related death using OpenSAFELY

Abstract

Coronavirus disease 2019 (COVID-19) has rapidly affected mortality worldwide¹. There is unprecedented urgency to understand who is most at risk of severe outcomes, and this requires new approaches for the timely analysis of large datasets. Working on behalf of NHS England, we created OpenSAFELY—a secure health analytics platform that covers 40% of all patients in England and holds patient data within the existing data centre of a major vendor of primary care electronic health records. Here we used OpenSAFELY to examine factors associated with COVID-19-related death. Primary care records of 17,278,392 adults were pseudonymously linked to 10,926 COVID-19-related deaths. COVID-19-related death was associated with: being male (hazard ratio (HR) 1.59 (95% confidence interval 1.53–1.65)); greater age and deprivation (both with a strong gradient); diabetes; severe asthma; and various other medical conditions. Compared with people of white ethnicity, Black and South Asian people were at higher risk, even after adjustment for other factors (HR 1.48 (1.29–1.69) and 1.45 (1.32–1.58), respectively). We have quantified a range of clinical factors associated with COVID-19-related death in one of the largest cohort studies on this topic so far. More patient records are rapidly being added to OpenSAFELY, we will update and extend our results regularly.

Our analyses provide a preliminary picture of how key demographic characteristics and a range of comorbidities—which were a priori selected as being of interest in COVID-19—are jointly associated with poor outcomes. These initial results may be used to inform the development of prognostic models. We caution against interpreting our estimates as causal effects. For example, the fully adjusted smoking

Policy implications and interpretation

The UK has a policy of recommending shielding (staying at home at all times and avoiding any face-to-face contact) for groups who are identified as being extremely vulnerable to COVID-19 on the basis of pre-existing medical conditions²⁵. We were able to evaluate the association between most of these conditions and death from COVID-19, and we confirmed the increased mortality risks, supporting the targeted use of additional protection measures for people in these groups. We have demonstrated that only a small part of the substantially increased risks of COVID-19-related death among BAME groups and among people living in more-deprived areas can be attributed to existing disease. Improved strategies to protect people in these groups are urgently needed²⁶. These might include the specific consideration of BAME groups in shielding guidelines and workplace policies. Studies are

<https://www.nature.com/articles/s41586-020-2521-4>



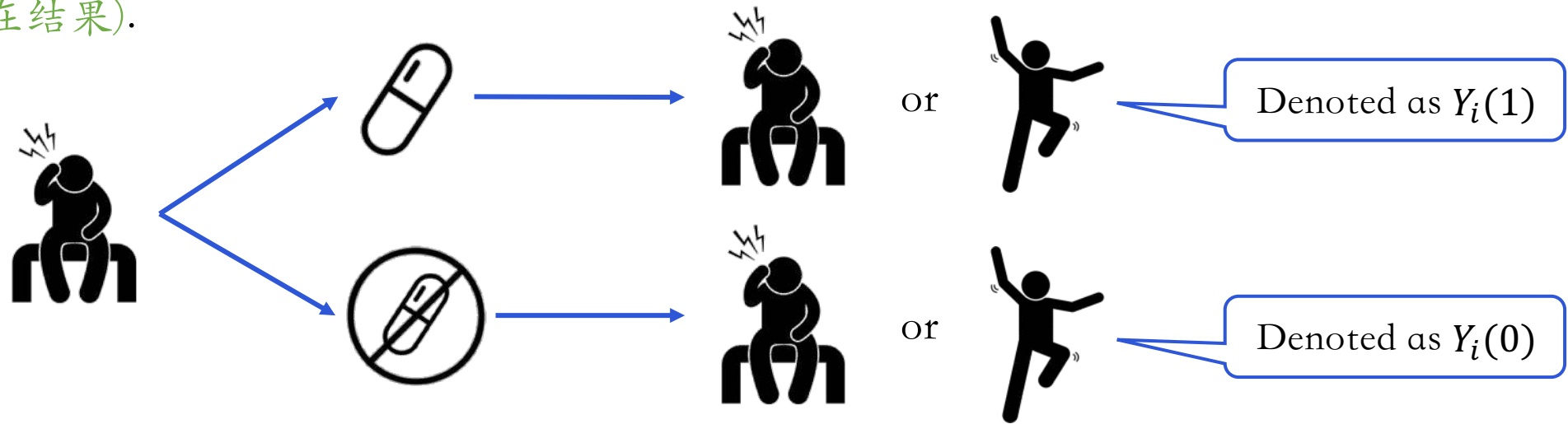
Chapter 8 Causal Inference

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8.2 Basic Concepts in Causal Inference

- Since correlation does not imply causation, what exactly is causation? How to express it mathematically?
- To formalize causality mathematically, we first introduce a key concept — **potential outcomes** (潜在结果).

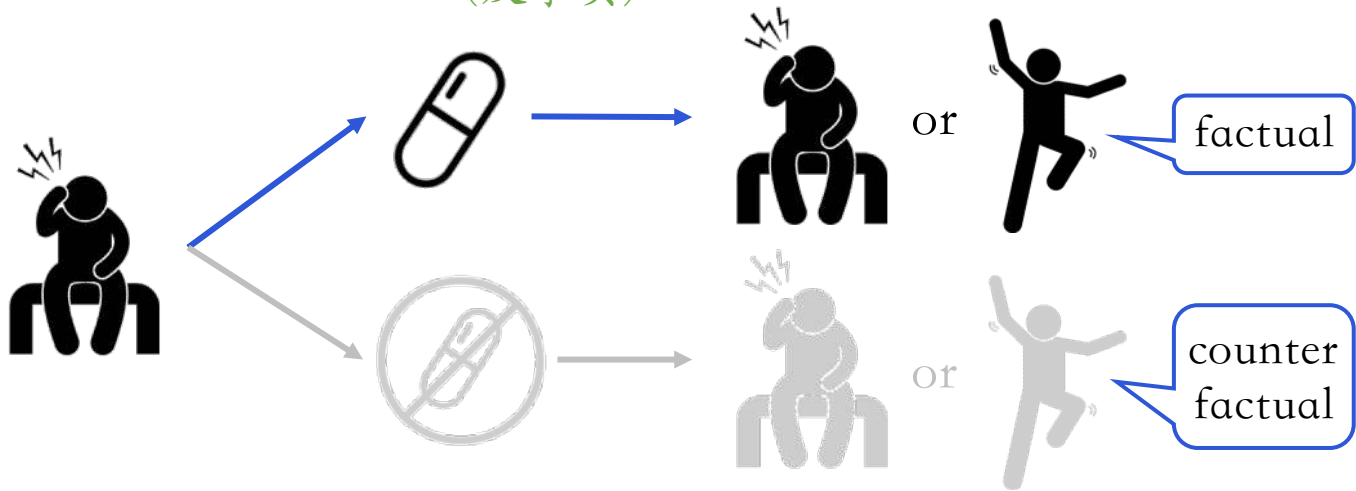


- $Y_i(1)$ and $Y_i(0)$ denote the potential outcomes for individual i under treatment (taking a drug) and control (not taking it), respectively.
- For a given individual i , the causal effect of the treatment is defined as $\tau_i \triangleq Y_i(1) - Y_i(0)$, which is called the **individual treatment effect** (ITE, 个体因果效应).



8.2 Basic Concepts in Causal Inference

- From the definition of potential outcomes, we see the fundamental problem of causal inference.
- For the same individual i , we can never observe both $Y_i(1)$ and $Y_i(0)$, making it impossible to measure the individual treatment effect τ_i .
 - The observed potential outcome is called the **factual** (事实), while the unobserved one is termed the **counterfactual** (反事实).



i	T	Y	$Y(1)$	$Y(0)$	$Y(1) - Y(0)$
1	0	0	?	0	?
2	1	1	1	?	?
3	1	0	0	?	?
4	0	0	?	0	?
5	0	1	?	1	?
6	1	1	1	?	?

- This issue can also be framed as a **missing data** (缺失数据) problem.
- Since the problem is inherently unsolvable at the individual level, causal inference shifts focus to the **average treatment effect** (ATE, 平均因果效应) $\tau = E[Y(1) - Y(0)] = E[Y(1)] - E[Y(0)]$.



8.2 Basic Concepts in Causal Inference

- How to estimate the average treatment effect?
- From the observed data provided, we can estimate the conditional expectations:

$$\hat{E}(Y|T = 1) = \frac{2}{3}, \quad \hat{E}(Y|T = 0) = \frac{1}{3}.$$

- Can we estimate $\tau = E[Y(1)] - E[Y(0)]$ with $\hat{E}(Y|T = 1) - \hat{E}(Y|T = 0) = 1/3$?

- The question is essentially asking

$$E[Y(1)] - E[Y(0)] \stackrel{?}{=} E[Y|T = 1] - E[Y|T = 0].$$

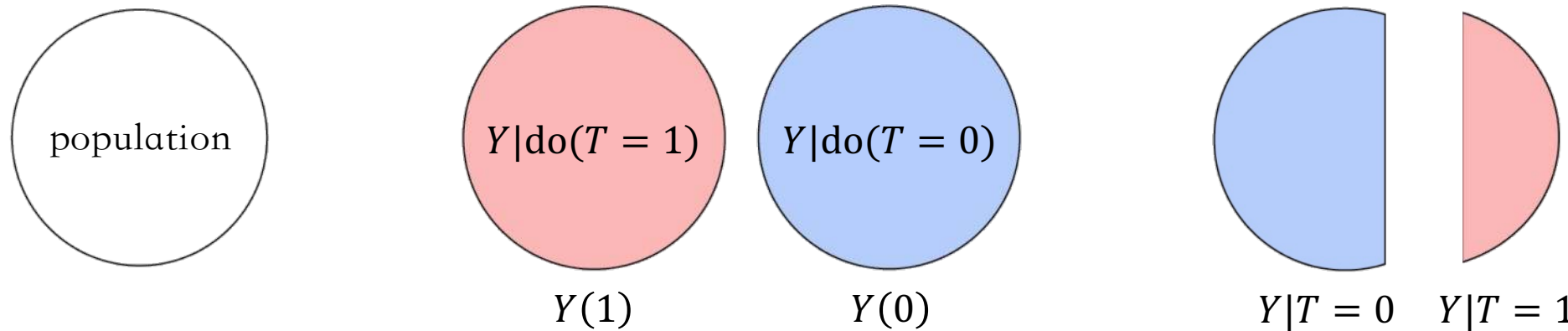
- This question has been answered by the statement “correlation does not imply causation”.
 - The left-hand-side of the equation is a **causal effect estimand** (因果效应被估量).
 - The right-hand-side of the equation is a **statistical effect estimand** (统计效应被估量).
 - Their meaning are essentially different.

i	T	Y	$Y(1)$	$Y(0)$	$Y(1) - Y(0)$
1	0	0	?	0	?
2	1	1	1	?	?
3	1	0	0	?	?
4	0	0	?	0	?
5	0	1	?	1	?
6	1	1	1	?	?



8.2 Basic Concepts in Causal Inference

- The figure below intuitively shows the difference between $E[Y(1)] - E[Y(0)]$ and $E[Y|T = 1] - E[Y|T = 0]$.

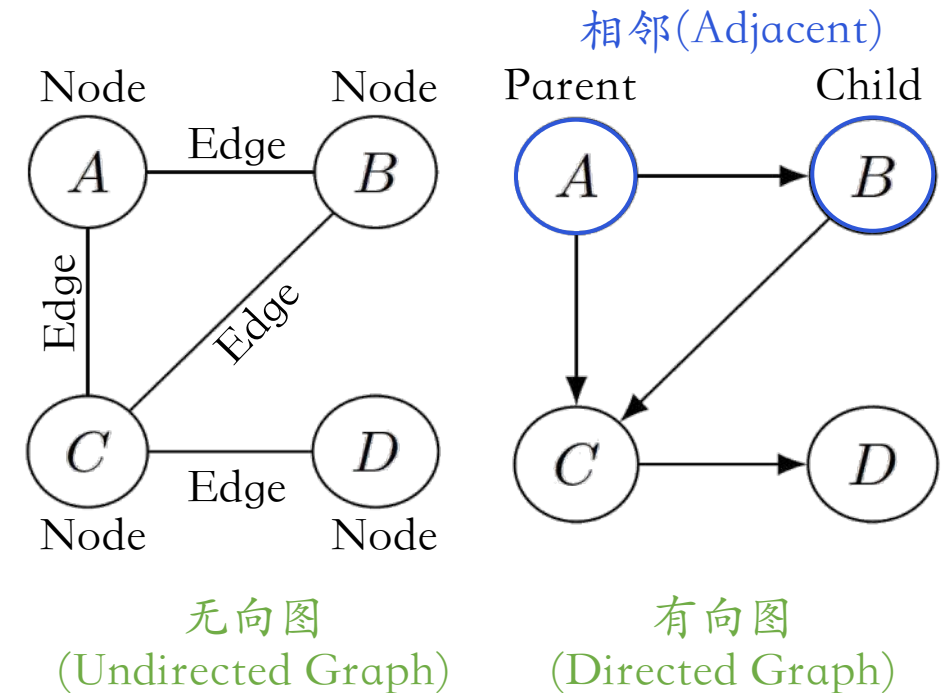
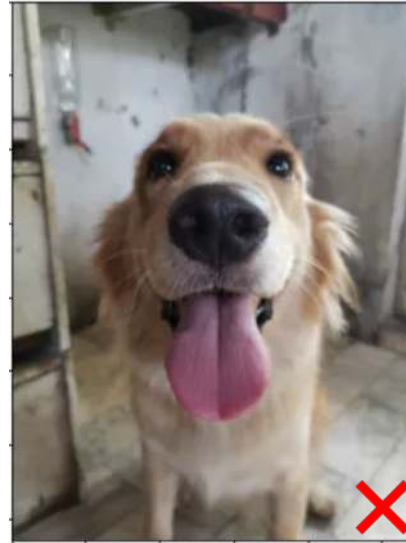
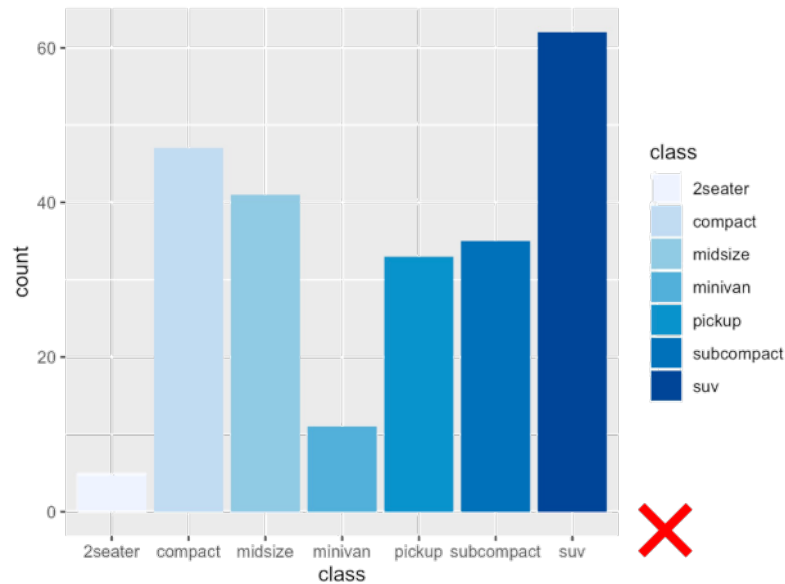


- $E[Y|T = 1] - E[Y|T = 0]$ can be obtained directly from the observed data, corresponding to the first level of the ladder of causation.
- $E[Y(1)] - E[Y(0)]$ belongs to the second level. To clearly express its difference with the first level, Pearl introduced the concept of **do operator** (do算子).
 - $P(Y|T = t)$ is the **conditional distribution** (条件分布) of the response variable Y under observation.
 - $P(Y|do(T = t))$ is the **interventional distribution** (干预分布) of Y under intervention.
 - Therefore, the average treatment effect can also be expressed as $E[Y|do(T = 1)] - E[Y|do(T = 0)]$.



8.2 Basic Concepts in Causal Inference

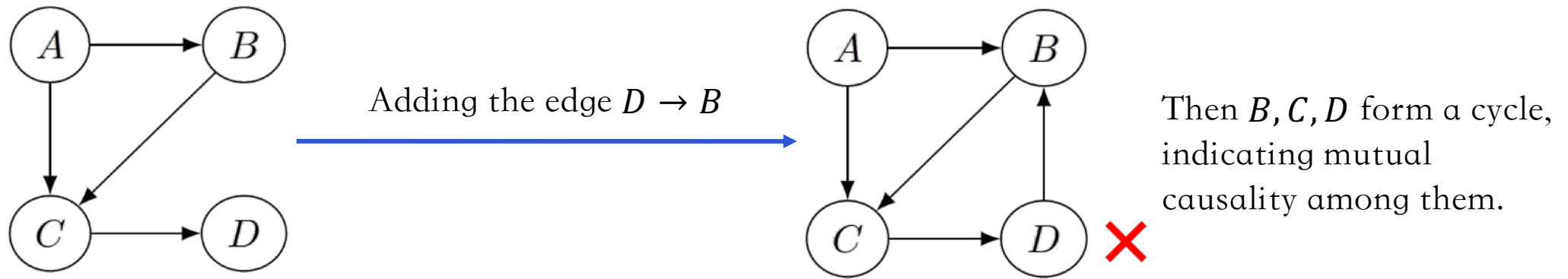
- Researchers often use **causal graph** (因果图) to visually express causal relationships.
- The graph here does not mean image or figure, but structure consists of **node** (节点) and **edge** (连边).



- Nodes in causal graph represent variables, directed edges represent causal relationships between variables, that is, change in the cause variable leads to **distribution change** of the result variable.



8.2 Basic Concepts in Causal Inference



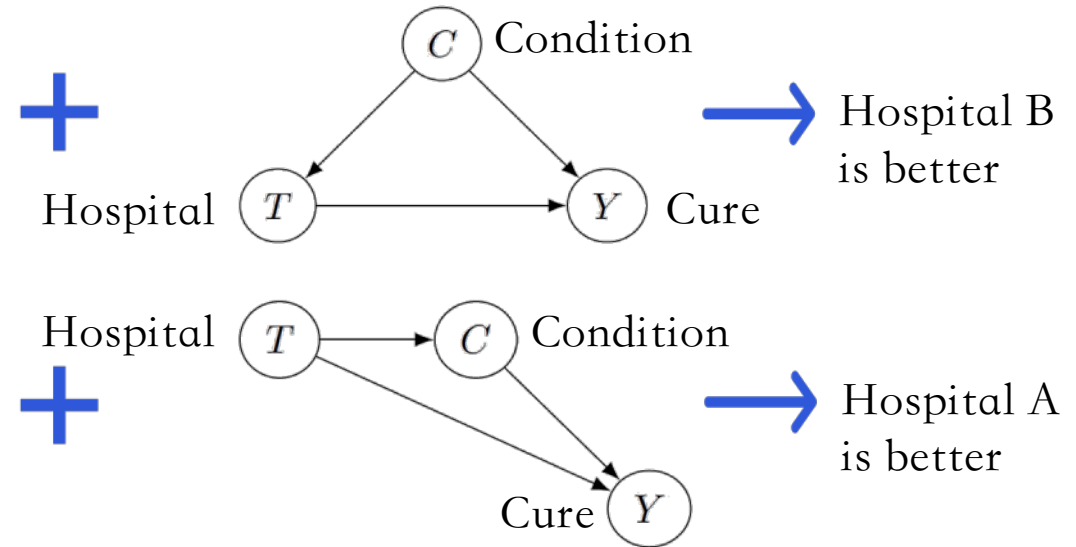
- Therefore, a causal graph must be a **directed acyclic graph (DAG, 有向无环图)**.
- If there exists a sequence of edges connecting two nodes, then we say there is a **path (路径)** between them.
- Both $A \rightarrow C \rightarrow D$ and $A \rightarrow C \leftarrow B$ are paths, i.e., edges in a path are not required to have the same direction.
- A path in a DAG never passes through the same node more than once.



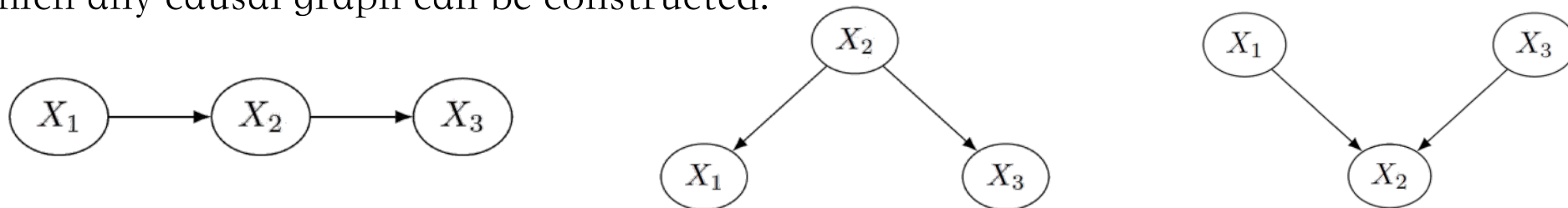
8.2 Basic Concepts in Causal Inference

- Based on observational data + causal graph, we can make better decisions. Take the Simpson's paradox as an example:

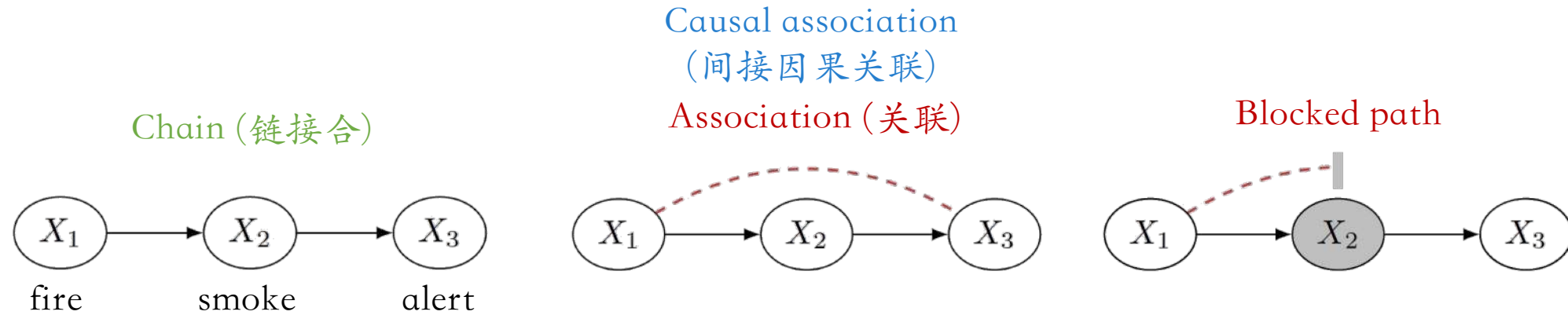
	Mild Case	Severe Case	Total
Hospital A	80% (720/900)	30% (30/100)	75% (750/1000)
Hospital B	90% (540/600)	40% (160/400)	70% (700/1000)



- The fundamental building blocks of all causal graphs are structures consisting of two edges and three nodes, known as **junctions** (接合). There are three basic types of junctions, from which any causal graph can be constructed.



8.2 Basic Concepts in Causal Inference



- X_2 is treated as a **mediator** (中介物), transmitting the effect of X_1 to X_3 .
- This setup implies the **conditional independence assumption**: given X_2 , X_1 and X_3 are independent.

Proof: The joint distribution of X_1, X_2, X_3 in a chain can be expressed as $P(X_1, X_2, X_3) = P(X_3|X_1, X_2)P(X_1, X_2) = P(X_1, X_2)P(X_3|X_2)$. Then, given X_2 , by the Bayes rule:

$$P(X_1, X_3|X_2) = \frac{P(X_1, X_2, X_3)}{P(X_2)} = \frac{P(X_1, X_2)P(X_3|X_2)}{P(X_2)} = P(X_1|X_2)P(X_3|X_2).$$

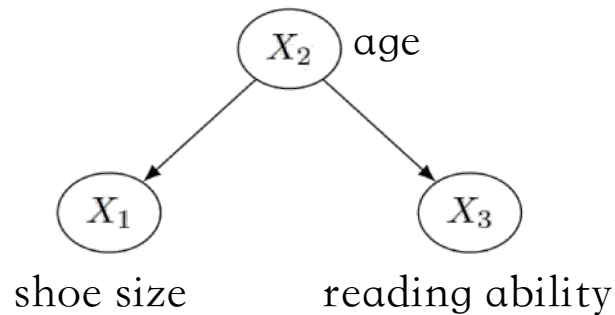
This suggests that $X_1 \perp X_3|X_2$.

- X_1 and X_3 exhibit an association in the observational data, but this association is **blocked** (切断) after controlling for X_2 .



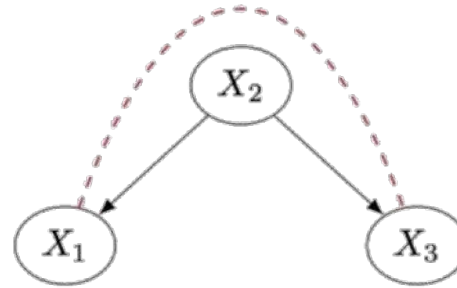
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Fork (叉接合)

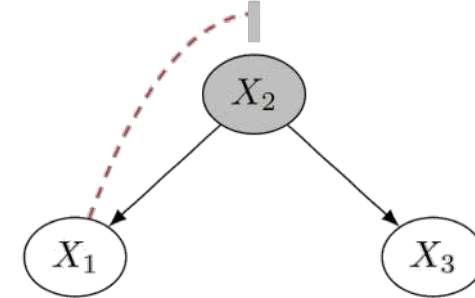


Non-causal association
(非因果关联)

Association (关联)



Blocked path



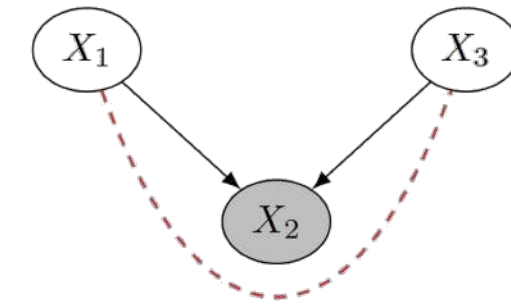
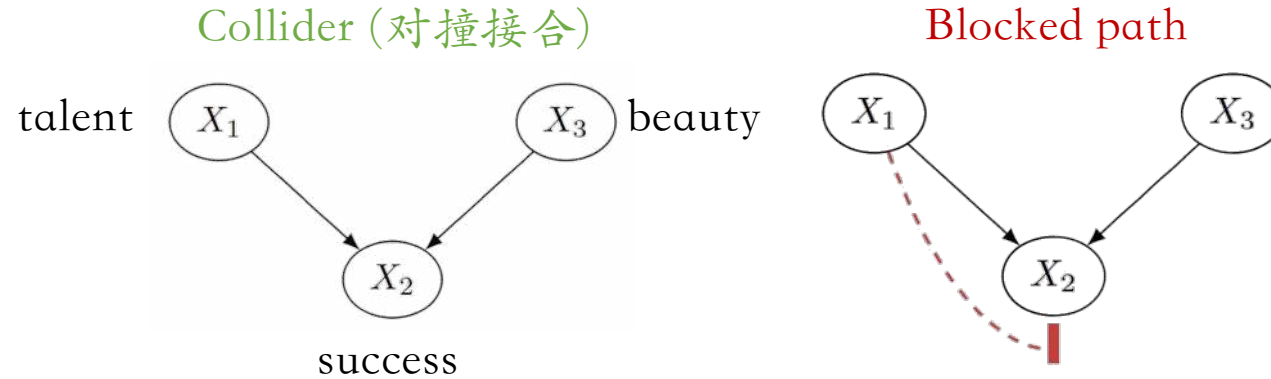
- X_2 is treated as the **common cause** (共因) or **confounder** (混杂因子) of X_1 and X_3 , it induce a statistical association between X_1 and X_3 , even in the absence of a causal relationship.
- As with a chain, X_1 and X_3 are conditionally independent given X_2 in a fork.
- Thus, by controlling for X_2 , we can eliminate the non-causal association between X_1 and X_3 , blocking the flow of information from X_1 to X_3 or vice versa.



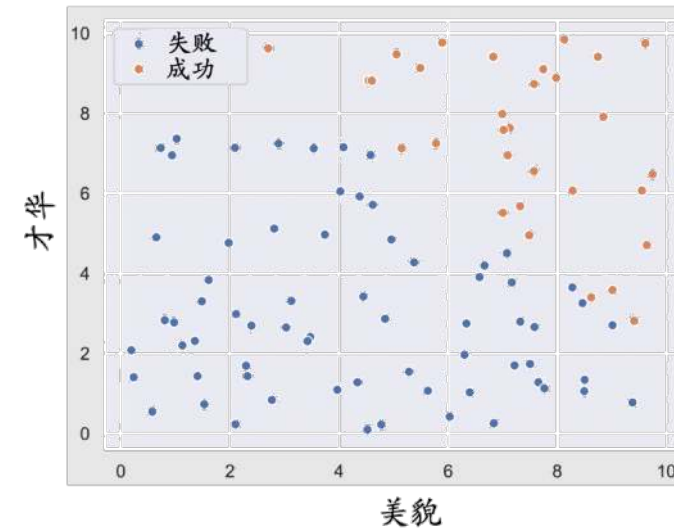
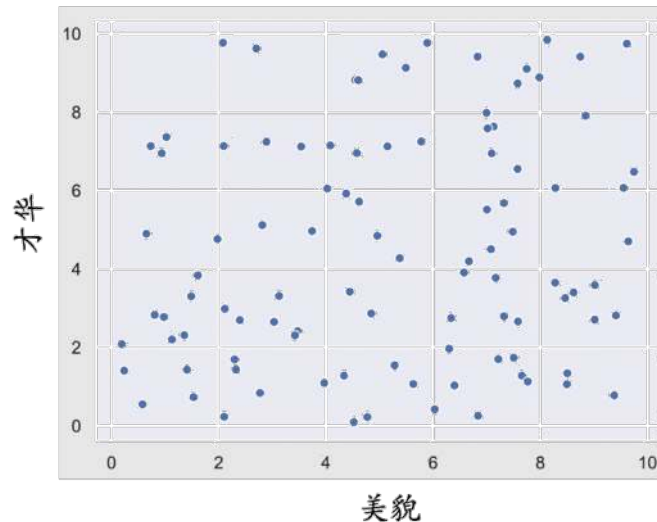
8.2 Basic Concepts in Causal Inference

Non-causal association
(非因果关联)

Association (关联)

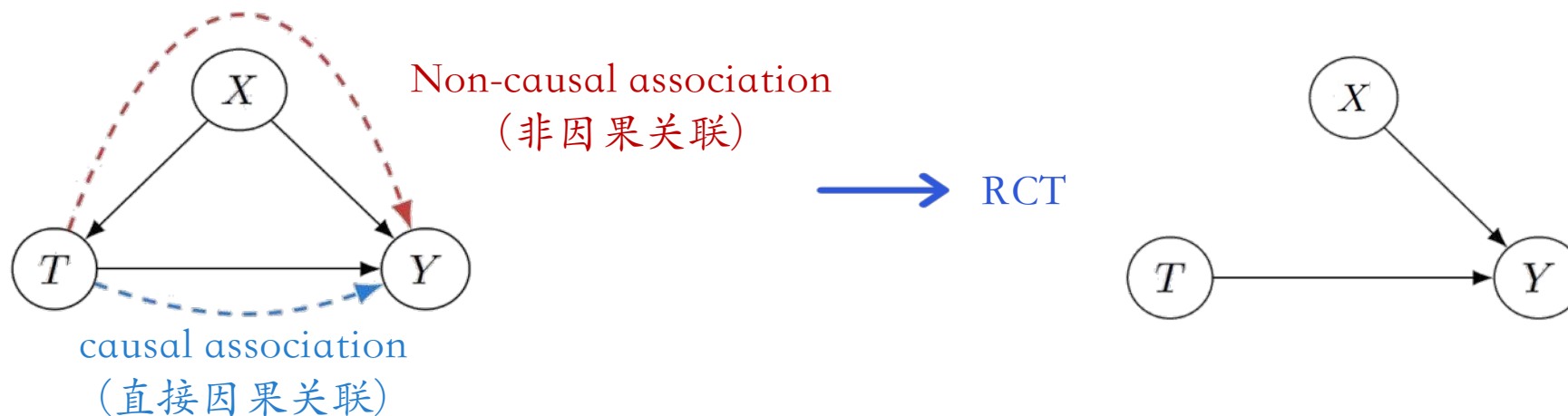


- Different from a chain/fork, X_1 and X_3 are independent in a collider, and conditioning on X_2 induces a non-causal association between X_1 and X_3 .



8.2 Basic Concepts in Causal Inference

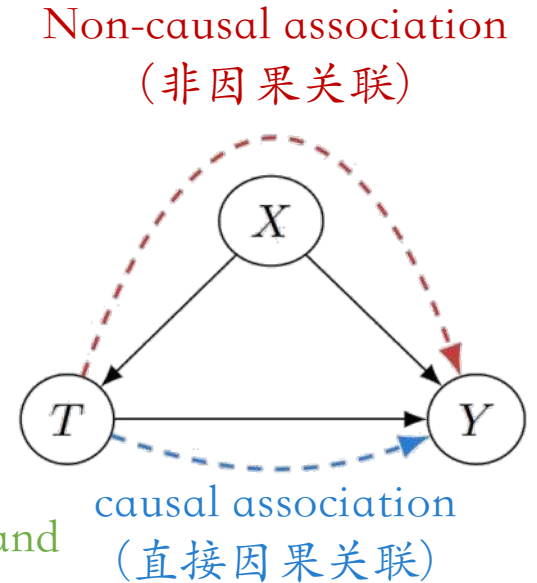
- To study the causal effect between two variables, we need to block the non-causal paths between them while keeping the causal paths open.
 - E.g., as the figure below shows, the association between T and Y is a mixture of casual and non-causal association.
 - To study the causal effect of T on Y , we need to block the path $T \leftarrow X \rightarrow Y$, called the **backdoor path** (后门路径).
- In an RCTs, we can actively intervene the value of T , i.e., $\text{do}(T)$, which eliminates all incoming arrows to T , thus blocking $T \leftarrow X \rightarrow Y$.



8.2 Basic Concepts in Causal Inference

- In observational studies, we can examine the association between T and Y conditional on X .
- As previously discussed regarding a fork, conditioning on X blocks the path $T \leftarrow X \rightarrow Y$.
 - If X is a categorical variable, we first estimate the causal effect of T on Y within each category of X , then compute the overall causal effect by taking a weighted average of these stratum-specific effects.

Causal effect estimand $\Rightarrow E[Y(t)] = \sum_{k=1}^K E[Y(t)|X = x_k] = E[Y(t)|T = t, X = x_k] = E(Y|T = t, X = x_k)$ Statistical effect estimand

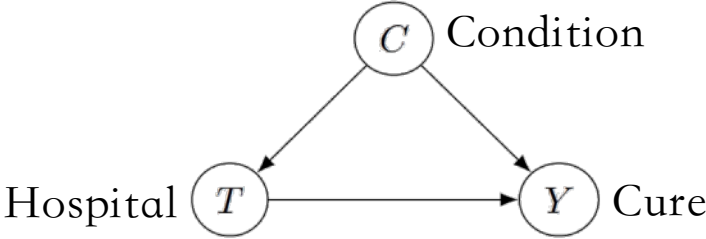


- If X is a numerical variable, similar formulation applies, with sum replaced by integration and probability replaced by density function.



8.2 Basic Concepts in Causal Inference

- Again, take the Simpson’s paradox as an example:



$$\hat{P}(C = 0) = \frac{1500}{2000} = \frac{3}{4}, \quad \hat{P}(C = 1) = \frac{500}{2000} = \frac{1}{4}.$$

	Mild Case	Severe Case	Total	Causal Effect
Hospital A	80% (720/900)	30% (30/100)	75% (750/1000)	67.5%
Hospital B	90% (540/600)	40% (160/400)	70% (700/1000)	77.5%

$\hat{E}(Y|T = t, C = 0)$
 $\hat{E}(Y|T = t, C = 1)$
 $\hat{E}(Y|T = t)$
 $\hat{E}[Y(t)]$

$$\hat{E}[Y(A)] = 80\% \times \frac{3}{4} + 30\% \times \frac{1}{4} = 67.5\%.$$

$$\hat{E}[Y(B)] = 90\% \times \frac{3}{4} + 40\% \times \frac{1}{4} = 77.5\%.$$

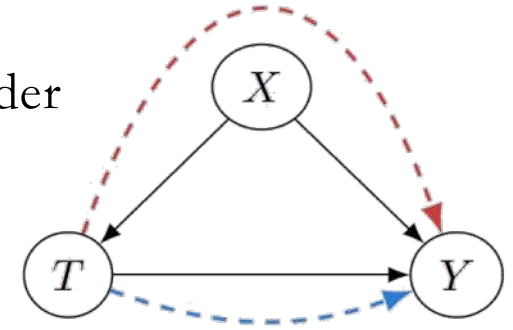


8.2 Basic Concepts in Causal Inference

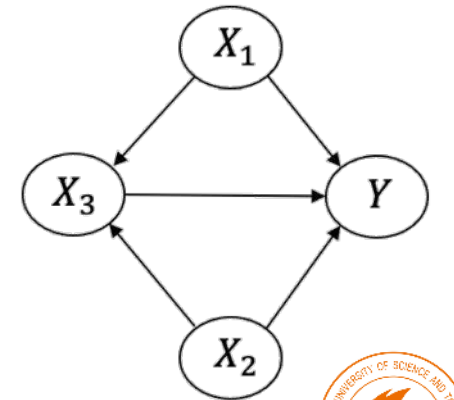
- In regression models, the back door path can be blocked by adding the confounder to the model as an explanatory variable. E.g.:

$$E(Y|T = t, X = x) = \beta_0 + \beta_1 t + \beta_2 x,$$

then β_1 is the causal effect of T on Y .

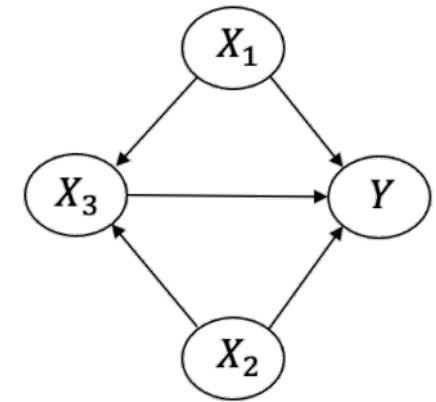


- However, without considering the causal relationships between variables, blindly adding more variables to a regression model is unreliable.
- Because some variables are not confounders but colliders. Controlling for colliders would inadvertently open backdoor paths.
 - As shown in the right causal graph, to study the causal effect of X_3 on Y , we need to close two backdoor paths: $X_3 \leftarrow X_1 \rightarrow Y$ and $X_3 \leftarrow X_2 \rightarrow Y$.
 - This requires fitting the regression model $Y \sim X_3 + X_1 + X_2$ rather than simply $Y \sim X_3$.



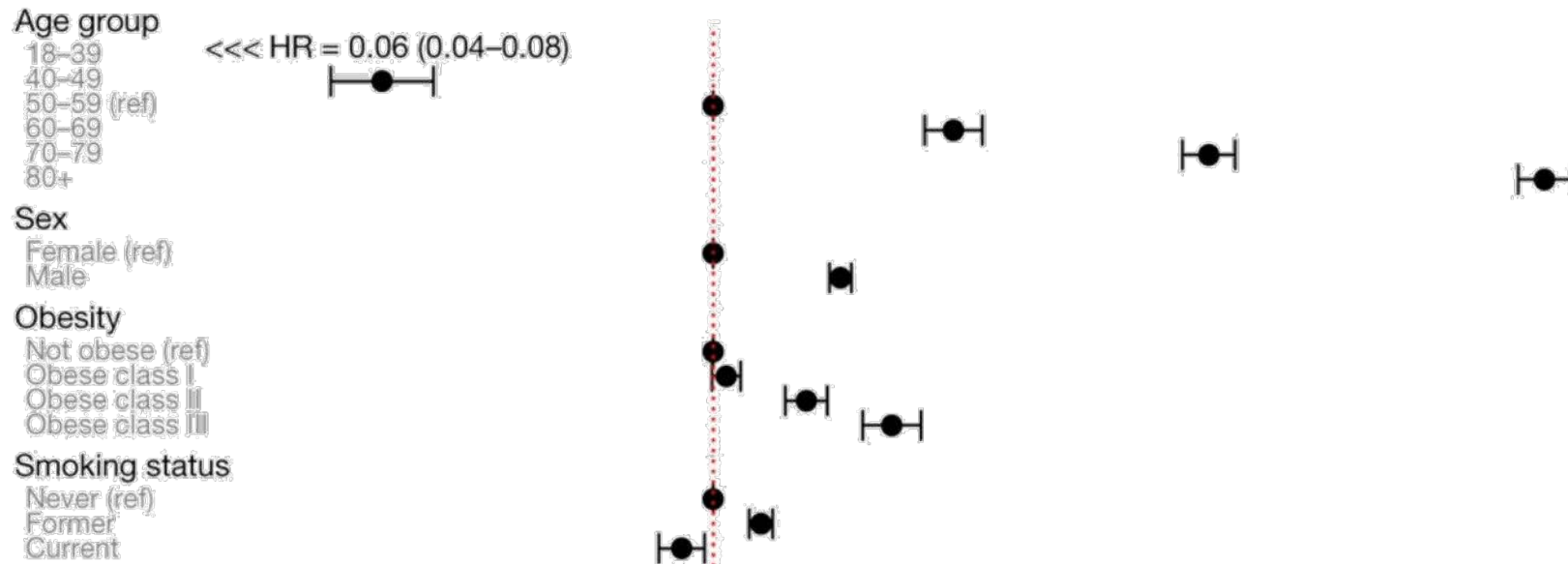
8.2 Basic Concepts in Causal Inference

- On the other hand, to examine the causal effect of X_1 on Y , there are three paths:
 - Direct causal effect: $X_1 \rightarrow Y$
 - Indirect causal effect: $X_1 \rightarrow X_3 \rightarrow Y$
 - Non-causal path: $X_1 \rightarrow X_3 \leftarrow X_2 \rightarrow Y$
- The third path is naturally blocked since X_3 is a collider.
- However, if we include X_3 in the regression model, this path would be opened.
- Therefore, to estimate the total causal effect (= direct effect + indirect effect) of X_1 on Y , we should simply fit $Y \sim X_1$ instead of $Y \sim X_1 + X_3$.
- In addition, fitting $Y \sim X_1 + X_3 + X_2$ would block both the second and third paths, yielding only the direct causal effect of X_1 on Y .
- In practical data analysis, researchers sometimes obtain counterintuitive results, which may occur when they fail to consider causal relationships between variables and simply include all available variables in the model.



8.2 Basic Concepts in Causal Inference

- As demonstrated in the paper discussed at the end of Section 7.1, factors associated with COVID-19-related death are given in the figure below.



- Most findings aligned with medical intuition, e.g., higher mortality risk for older people, greater risk for males than females, increased risk with higher obesity levels, people with smoking history showed higher risk than never-smokers.
- Yet paradoxically, current smokers appeared to have reduced risk.

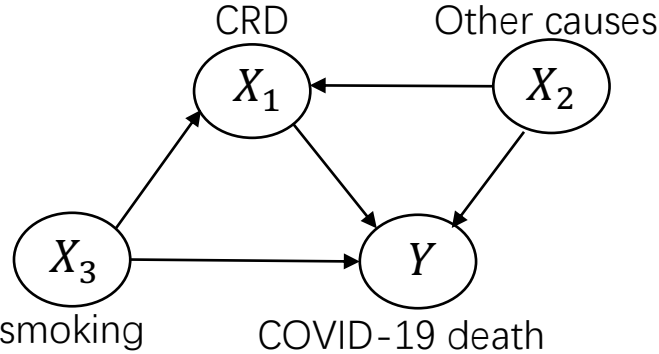


8.2 Basic Concepts in Causal Inference

- The authors found that after adding **Chronic respiratory disease (CRD)** to the regression model, the hazard ratio for smoking reversed direction.

Smoking	HR without CRD	HR with CRD
Never	1.00	1.00
Former	1.43 (1.37-1.49)	1.19 (1.14-1.24)
Current	1.14 (1.05-1.23)	0.89 (0.82-0.97)

- This phenomenon becomes explicable after considering the causal graph.



Post hoc analyses of smoking and hypertension

Both current and former smoking were associated with a higher risk in models that were adjusted for age and sex only, but in the fully adjusted model current smoking was associated with a lower risk (fully adjusted HR 0.89 (0.82–0.97)), which concurs with the lower than expected prevalence of smoking that was observed in previous studies among patients with COVID-19 in China¹⁰, France¹¹ and the United States¹⁹. We investigated this in more depth post hoc by adding covariates individually to the age, sex and smoking model, and found that the change in hazard ratio was driven largely by adjustment for chronic respiratory disease (HR 0.98 (0.90–1.06) after adjustment). This and other comorbidities could be consequences of smoking, highlighting that the fully adjusted smoking hazard ratio cannot be interpreted causally owing to the inclusion of factors that are likely to mediate smoking effects. We

- The path $X_3 \rightarrow X_1 \leftarrow X_2 \rightarrow Y$ is originally blocked, but conditioning on X_1 opens this non-causal association path.



